A Better Hand

*The PIE contrastive *-tero- and the pre-PIE hand-noun*

Paper 2 of the der- project. Builds on Chavis & Claude (2026a), "The Hand That Gives" (LingBuzz v23).

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Abstract

The PIE contrastive suffix *-*tero*- is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Almost every PIE local particle has a *-*tero*- derivative naming its spatial contrastive — Latin *citra/ultra/intra/extra/contra*, Sanskrit *ádhara-/uttara-/ápara-/ántara-*, Avestan and Old Persian *fra-tara-/apa-tara-/upa-tara-*, Hittite *kattera-*, Greek *próteros/hústeros*. The Handbook of Comparative and Historical Indo-European Linguistics states explicitly that outside Greek and Indo-Iranian, **-tero-* and its bare alternant *-*(e)ro*- are confined to derivations from pronouns and particles; the gradational "more X" comparative function is a Greek and Indo-Iranian innovation.

This paper proposes that the original spatial-direction function of *-*tero*- follows naturally from the framework of Paper 1: pre-PIE **der-* "hand, open palm" grammaticalized through the pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and the resulting suffix was applied productively to local particles. The mainstream consensus that *-*tero*- originally marked bilateral contrast — established in every standard reference (Sihler 1995; Fortson 2010) — is what the framework is trying to explain. The *ambidexter* compound makes the bilateral presupposition explicit: it can only mean "having the right-side marker on both sides" if *-*tero*- normally marks one of two. The framework explains where that bilateral norm came from; *ambidexter* shows it was there.

The paper makes one positive empirical contribution beyond the etymological argument. The reversal hypothesis predicts that bare *-*(e)ro*- forms should cluster on stop-final bases where *-*tero*- would create OCP-violating clusters, while t-bearing forms cluster on bases ending in vowels, sonorants, and fricatives. Run on the inherited PIE positional adjective paradigm in Sanskrit, Avestan, Latin, and Greek, this prediction is matched in approximately 19 of 19 etymologically transparent cases. The test argues against the older "irregular extension" framing of Szemerényi and Sihler; it does not by itself decide between the framework's reversal and Kahl's (2024) metanalysis hypothesis. The framework's advantage over Kahl is not the distributional data but the etymological motivation: the framework explains why the *t*- was there from the start, as the initial consonant of the hand-noun **der-*, devoiced in suffix-initial position when the noun grammaticalized into a bound suffix. Kahl's metanalysis leaves the source of the *t*- open.

1. Introduction

The *-*tero*- suffix has been thoroughly documented in the Indo-European literature, but it has never been satisfactorily explained. The standard handbooks record that it marks bilateral contrast in its oldest function, that it applies systematically to local particles across the family, and that the gradational comparative is a later innovation in Greek and Indo-Iranian. What they do not explain is where the bilateral frame came from, or what the suffix was before it was a contrastive morpheme.

This paper proposes the missing etymology. Pre-PIE **der-* — the same body-part word for "hand, open palm" that Paper 1 (Chavis & Claude 2026a) reconstructed on the basis of Greek *δῶρον*, Hesychian *δάρις*, Old Irish *dorn*, Welsh *dwrn*, and Latvian *dūre* — grammaticalized through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE and became the suffix the handbooks reconstruct as *-*tero*-. The spatial-direction function preserved across the family is what remains of the hand-noun's body-part semantics after the grammaticalization ran its course.

A note on the *t/d* alternation before proceeding. Earlier versions of this project wrote the pre-PIE etymon as **t/d-er-* — a notational hedge reflecting the *t-* of the suffix alongside the *d-* of the lexical cognates. That hedge is retired here. The root is **der-* with original *d-*, following the consistent *d-*-vocalism of Greek *δῶρον*, Irish *dorn*, Welsh *dwrn*, and Latvian *dūre*. The *t-* of the grammaticalized suffix reflects regular suffix-initial devoicing of *d-* in the PIE morphological system: PIE bound suffixes show a strong preference for voiceless obstruents in initial position (*-*to-*, *-*ter-/-tor-*, *-*tro-*, *-*ti-*, *-*tu-* are all voiceless-initial). When **de-* grammaticalized into a bound suffix, its initial *d-* devoiced to *t-* in conformity with this pattern — the regular consequence of suffix-initial position, not a new sound rule.

The paper proceeds as follows. §2 documents the spatial-direction paradigm across the IE branches — the empirical center, presented in Table 1. §3 develops the bilateral argument: why *-*tero-* marks one-of-two rather than one-of-many, and what *ambidexter* shows. §4 engages the directionality question and Kahl (2024), the most current dissertation-length treatment of these morphemes. §§5–6 present the OCP mechanism and anchor it in the Hittite test case. §7 runs the distributional test. §8 develops the Anatolian distribution argument. §9 takes up the typological question. §10 flags the two broader extensions of the framework, with a reference to Paper 3. §§11–12 record future research and remaining problems.

2. The spatial-direction paradigm

We begin with the empirical center. Table 1 sets out the *-*tero-* paradigm across the IE branches: every major local particle, the *-*tero-* derivative it forms, and the spatial-direction adjective that results. The entries are restricted to inherited PIE-level formations; post-PIE extensions are discussed separately in §7.

Table 1. *The PIE *-tero- spatial-direction paradigm across branches.*

PIE particle	Meaning	Branch	*-tero- form	Derived meaning
*ḱi-	“this, here”	Latin	_citer, citra_	“nearer, to this side”
*ol- / uls-	“that, beyond”	Latin	_ulterior, ultra_	“farther, beyond”
*ud-	“up”	Sanskrit	_uttara-_	“upper, northern”
*ud- / uds-	“up, away”	Greek	_hústeros_	“later, after”
*ádh- / n̥dh-	“below”	Sanskrit	_ádhara-_	“lower”
*ádh-	“below”	Avestan	_aδara-_	“lower”
*n̥dh-	“below”	Latin	_inferus_	“lower” (bare alternant)
*(s)up- / uper-	“above”	Latin	_superus, super_	“upper, above”
*(s)up-	“above”	Avestan	_upara-, upataro-_	“upper, on top of”
*(s)up-	“above”	Greek	_hupér_	“above” (bare alternant)
*ḱm̥t-	“below/alongside”	Hittite	_kattera-_	“lower, inferior”
*$h_1$en-	“in”	Latin	_inter, intra_	“between, within”
*$h_1$en-	“in”	Sanskrit	_ántara-_	“inner, between”
*$h_1$en-	“in”	Greek	_éntera_ (n.pl.)	“innards” (lexicalized)
*eks-	“out”	Latin	_exter, extra_	“outer, outside”
*kom-	“with”	Latin	_contra_	“opposite, against”
*post-	“after”	Latin	_posterus_	“later, behind”
*pro-	“forward”	Avestan/OP	_fratara-_	“forward, more forward”
*pro-	“forward”	Greek	_próteros_	“former, before”
*ápa-	“away”	Sanskrit	_ápara-_	“posterior, the other”
*ápa-	“away”	Avestan	_apataram_	“farther away”
*deḳs-	“right”	Latin	_dexter_	“right-handed”

**deḳs-*	“right”	Greek	*deksiterós*	“right (hand)”
**swíþ-*	“strong, right”	Old English	*swiþro*	“right (hand or side)”
*k^w*o-*	“who”	Latin	*uter*	“which of two”
*k^w*o-*	“who”	Sanskrit	*katára-*	“which of two”
*k^w*o-*	“who”	Avestan	*katāra-*	“which of two”
*k^w*o-*	“who”	Greek	*póteros*	“which of two”
**al-*	“other”	Latin	*alter*	“the other (of two)”
**anyá-*	“other”	Sanskrit	*anyatara-*	“the other (of two)”
(deictic)	“other”	Greek	*héteros*	“the other”

The paradigm in brief

Latin preserves the most extensive paradigm. The bare alternant forms in *-trā* and *-trō* give the lexicalized adverbs *citra, ultra, intra, extra, contra* (de Vaan 2008). Latin *matertera* “maternal aunt” (de Vaan 2008: 369) — etymologically “the analogous mother (of two)” — is a Latin-internal extension of **-tero-* to a kinship term, noted here as it touches the kinship-suffix question left for §11.

Sanskrit preserves a closed paradigm of place-directional adverbs in *-tra* (the bare alternant): *átra* “here,” *tátra* “there,” *yátra* “where,” *kútra* “where (interrogative).” The lexicalized substantive *āntrá-* “entrails” (the neuter of *ántara-* “inner”) is worth marking: Latin *intestina*, Greek *éntera*, and Sanskrit *āntrá-* all independently lexicalize the neuter of **$h_1$en-tero-* as “the inner parts” — three branches converging on the same lexicalization confirm the PIE-level age of the spatial adjective and its body-part orientation.

Avestan and Old Persian preserve the most archaic system. Milizia (2020: 268, 271) observes that the only Old Persian documented *-tara-/-tama-* formations are derivatives of local particles or preverbs. There are no gradational comparatives in *-tara-* in Old Persian. This is not a corpus accident; it reflects, on Milizia’s reading, the most archaic preserved layer of the formation. The expansion to gradational comparison is attested in Greek and Indo-Iranian (and partially elsewhere) but absent from the most archaic Old Persian layer.

Greek attests both the spatial-direction paradigm (Table 1) and, innovatively, the gradational “more X” function (*sophúteros* “wiser”). **Hittite** preserves only one **-tero-/*-(e)ro-* form productively: *kattera-* “lower; inferior; further along” (Kloekhorst 2008: 538–539), built on the “below”-particle *katt-*. The productive contrastive function in Anatolian is taken over by **-eti-Ho-* (*-ezziya-*), in *ḫantezziya-* “first, foremost” and *appezziya-* “back, last.” We return to the significance of this asymmetry in §8.

The handbook’s verdict. The Handbook of Comparative and Historical Indo-European Linguistics (vol. 3) is explicit: outside the Greek and Indo-Iranian extension into gradational comparison, **-tero-* and **-(e)ro-* “are confined to derivations of pronouns and particles.” This is not the framework’s claim; it is the standard view. The original PIE function of the suffix was exactly the paradigm documented in Table 1. The framework’s contribution is an etymological account of why that particular suffix should perform that particular function — something the standard treatment leaves open.

3. The bilateral character of **-tero-* and the *ambidexter* argument

The standard handbooks agree on one thing about **-tero-* that the framework is trying to explain: the suffix’s original function was bilateral contrast — not general gradation, not loose comparison, but specifically “one of a pair as opposed to the other.” Fortson’s formulation is precise: the basic function of the suffix was to “set off a member of a pair contrastively from the other member of the pair, as in Greek *skaiós* ‘left’ vs. *deksiterós* ‘right,’ i.e. ‘the one on the right (as opposed to the one on the left)'" (Fortson 2010). Sihler (1995: 360–361) gives the same analysis. The canonical examples — *póteros* "which of two" and *deksiterós* "right (as opposed to left)" — confirm it. The bilateral character of *-*tero*- is the mainstream consensus. What the mainstream does not provide is any account of where that bilateral character comes from. It is treated as a primitive property of the suffix, with no etymology.

The framework answers that question. Bare **de*- — the directional particle documented in Paper 3 of this project (Chavis & Claude 2026c) as the morpheme underlying the Greek allative -*δε*, Old Church Slavonic *do*, and English *to* — is an absolute directional: "toward there," pointing at a target with no implied alternative and no bilateral frame. Greek *oíkade* "homeward" does not presuppose a contrasted alternative direction. The locative *-*r*- element (Lipp 2009; Paper 3 §3) changes this. It anchors the directional to the body as frame of reference — "at the place of the hand-direction" — and bodies have inherent bilateral symmetry. The hand is the body's most salient bilateral pair: exactly two, one on each side, symmetrically opposed. When **de*- is grounded by locative *-*r*- into a body-anchored location, the result is inherently bilateral, because the question "which side of the body?" applied to an organism with two hands has exactly two answers. The absolute directional becomes a relational one.

The bilateral character was established in the suffix's earliest productive use — with **deḱs*- "right," giving **deḱs-i-tero*- "right as opposed to left," where the two hands provided the bilateral frame. Once that function was established through the first productive context, it grammaticalized into the suffix and traveled with it to other paired domains: up/down (*uttara-/ádhara*-), before/after (*próteros/hústeros*), inner/outer (*ántara-/exter*). The hand etymology explains the origin of the bilateral character; the later extensions explain its spread. Paper 3 §4.2 completes the compositional argument, showing that the bilateral frame is a structural consequence of what **de-r-o*- composes to mean, not a grammaticalized accident.

Ambidexter is the compound that makes this argument explicit. It parses as *ambi*- (PIE *h_2mb^hi "on both sides") + **deḱs*- ("right") + -*i*- (linking vowel) + *-*tero*- + Latin nominal ending: "having the right-side marker on both sides." That statement is only meaningful if *-*tero*- normally marks one of two bilateral sides. You can only say "on both sides" if the default is "on one side." The compound's structure — *ambi*- negating a bilateral norm — presupposes and thereby reveals the bilateral character the framework derives from the hand-noun. *Ambidexter* is not evidence that *-*tero*- is a hand-noun; it is evidence that *-*tero*- is a bilateral-contrast marker whose bilateral character the framework explains and the standard treatment leaves open. Festus's gloss confirms the parse: *ambidexter qui dexteros ambo manus habet* — an ancient recognition that *ambi*- governs the compound and that *-*tero*- normally picks out one of two. The Greek cognate *deksiterós*, inherited independently from PIE, confirms that the bilateral frame is not a Latin formation. It is PIE.

4. The directionality question

4.1 The standard view

The standard view, summarized by Szemerényi (1996: 197), holds that the bare suffix *-*(e)ro*- is original and the t-bearing *-*tero*- is a secondary extension. This is the view of Wackernagel & Debrunner (1957) for Sanskrit, Schwyzer (1939: 533) for Greek, and Sihler (1995: 360–361) and Weiss (2009: 357–358) for the comparative treatment generally.

The most current and detailed elaboration is Kahl (2024), the recent Harvard dissertation on Indo-European degree morphology. Kahl's account is more sophisticated than the older "t-extension" framings: he proposes that bare *-*(e)ro*- has a locatival origin, that *-*ter* and *-*tero*- arose by metanalysis from forms with t-final stems being reanalyzed (the t-final of the stem reinterpreted as belonging to a new suffix *-*ter*/*-*tero*-), and that the path from emphatic to comparative was a subsequent development.

4.2 Where the framework agrees with Kahl

Kahl's §2.3.1 argues that bare *-*(e)ro-*/-*er* has a locatival origin — that the suffix marked spatial location of the kind familiar from Indo-European positional adjectives. This is independent published support for a key piece of the framework. The framework reads *-*tero-* as **de-* (hand, directional) + *-*r-* (locative) + *-*o-* (thematic). Kahl reads bare *-*(e)ro-* as a primitive locatival morpheme. The two readings agree that the *r* of *-*tero-* is locatival in origin. They are the same morpheme. Paper 3 develops this compositional analysis in detail.

4.3 Where the framework disagrees with Kahl

Where Kahl reads the *t* in *-*ter*/*-*tero-* as metanalyzed in from t-final stems, the framework reads it as the initial consonant of the hand-noun **de-* — devoiced in suffix-initial position, as explained in §1. Kahl's metanalysis requires an additional step: speakers reinterpreting the stem-final *t* of forms like **X-t-er-* as belonging to a new suffix *-*ter-*/*-*tero-*, then generalizing that suffix to bases that never had the *t*. This is a respectable mechanism, but it leaves open why the *t* specifically (rather than some other consonant) got reanalyzed, and why the resulting suffix had spatial-contrastive semantics from the start. The framework answers both: the *t* was the hand-noun's initial, and the spatial-contrastive semantics are the hand-noun's body-part content.

Both Kahl's account and the framework can in principle accommodate the same distributional facts (§7). The two hypotheses are not distinguished by the distributional data but by their etymological commitments. Kahl leaves the source of the *t-* open. The framework fills it.

4.4 What would distinguish the hypotheses

The framework's reversal predicts that bare *-*(e)ro-* forms arise by OCP-driven cluster simplification at morpheme boundaries, with deletion of the suffix-initial *t* of *-*tero-*. This predicts that bare forms should cluster on stop-final bases, and t-bearing forms should cluster on non-stop bases. Under Kahl's metanalysis, the same distribution follows from the source-form environments (t-final stems, which would naturally not include stop-final bases). Both hypotheses therefore predict approximately the same surface distribution.

What the distributional test in §7 does decide against is the older, looser version of the standard view — in which the t-extension is treated as analogically irregular, with no systematic phonological conditioning. The Szemerényi/Sihler framing leaves the distribution as a lexical accident. The data argue against that framing. Whichever account of the *t* — reversal or metanalysis — turns out to be right, the "irregular extension" version is what the inherited paradigm's systematic conditioning rules out.

5. The reversal: existing phonological rules deliver the bare form

The phonological mechanism by which an original *-*tero-* could yield bare *-*(e)ro-* in some environments does not require a new sound rule. The mechanism is cluster simplification at morpheme boundaries, driven by the Obligatory Contour Principle (OCP), as documented for PIE in Byrd (2015).

Byrd's framework is the current authoritative treatment of PIE syllable structure. The OCP — the principle that PIE banned geminates across morpheme boundaries — is documented as undominated in PIE following Meillet (1934). When morpheme-boundary clusters violated OCP, PIE had two resolution strategies: s-epenthesis for VTTV sequences (**úeid-tó-* > **úeit.stó-*), and consonant deletion for VTTRV sequences where epenthesis would create an illegal coda (**méd-trom* > Greek *métron*). The latter is the Métron Rule (Saussure 1885; Mayrhofer 1986: 111; Byrd 2015: ch. 3.6).

The Métron Rule is exactly the mechanism we need. It deletes a dental in dental–dental–resonant sequences at morpheme boundaries, yielding a single dental in the surface form. It is well established as a PIE process. Crucially, Byrd is explicit that the rule's exact deletion target is genuinely undecided: his footnote 161 reads "If the /d/ of the root was lost, we would expect *mé.trom*; if the /t/ of the suffix was lost, we would expect *mét.rom* (see chapter 7 for discussion)." Greek *métron* with single *-t-* is consistent with both outcomes. The Métron Rule therefore has two alternative outcomes; the specific case does not disambiguate.

We propose that the same rule applies to the case at hand: pre-PIE **X-tero-* (where X is a stem ending in a dental, as in **ḱm̥t-tero-*) is reduced to **X-(e)ro-* by deletion of the suffix-initial dental. The rule is not new. The deletion target is the alternative one Byrd himself explicitly recognizes as a valid possibility. The Hittite contrastive *kattera-*, the empirical anchor of §6, picks the suffix-initial deletion over the root-final deletion — making Byrd's footnote 161 empirically decidable.

6. The Hittite test case: *kattera-*

The Hittite contrastive adjective *kattera-* "lower; inferior; infernal; further along" (Kloekhorst 2008: 538–539) is the empirical anchor for the phonological argument. It is built on the "below"-particle **ḱm̥t-* that gives Latin *cum*, Old Irish *cét*, Greek *katá*, Hittite *katta*.

The derivation under the framework runs in four steps:

pre-PIE: **ḱm̥t-tero-*	*root + suffix; dental cluster across morpheme boundary*
PIE: **ḱm̥t-ero-*	*OCP / Métron Rule; suffix-initial t deleted*
Anatolian: **katt-era-*	**m̥ vocalizes to a; *m before *t assimilates to geminate*
Hittite: *kattera-*	*regular Anatolian outcome (Melchert 2017: 11–12)*

The alternative deletion target — root-final *t* — would have predicted Hittite *katera-* with a single intervocalic *-t-*, not the attested geminate *-tt-* of *kattera-*. The Hittite outcome picks the suffix-initial deletion. This is the empirical bite of Byrd's footnote 161: there are surface forms that disambiguate the rule's outcome, and the Hittite contrastive is one of them.

Lipp's (2017) semantic analysis of **ḱm̥t-* is worth noting. In his GeSuS abstract on Hittite *katta* and the Luvo-Lycian dorsal palatalization, Lipp glosses pre-PIE **ḱm̥t-* not merely as "below" but more precisely as "into close contact, into/under pressure" — a reading that bridges the apparently divergent senses "alongside" (preserved in Latin *cum*, Greek *katá* in some uses) and "downwards" (Hittite *katta*, *kattera-*) under a single spatial concept of close contact-from-below. Under Lipp's gloss, the contrastive **ḱm̥t-tero-* "the alongside-or-downward-side one (of two)" parses with greater semantic transparency than the simpler "below"-gloss alone permits.

Both Kloekhorst's analysis and the reversal predict the same surface form. The data alone do not distinguish them. Two considerations weigh in favor of the t-bearing analysis. First, etymological economy: if pre-PIE **der-* was a body-part word, then the t-bearing suffix is the historically prior form, and the bare form requires a phonological explanation — which the Métron Rule supplies. Second, paradigm-level distribution: the t-bearing alternant is productive across the entire spatial-direction paradigm documented in §2, while the bare alternant is preserved only in lexicalized relics. This is the distribution expected if the t-bearing form is original.

7. The distributional test of the reversal hypothesis

7.1 The prediction

Under the reversal hypothesis, bare **-(e)ro-* forms arise by OCP-driven cluster simplification: the suffix-initial *t* of **-tero-* is deleted when the base ends in a stop, creating a stop+stop cluster that violates the OCP. The reversal hypothesis therefore predicts that bare **-(e)ro-* forms should cluster systematically on stop-final bases, while t-bearing **-tero-* forms should cluster on bases ending in vowels, sonorants, or fricatives (where no OCP violation arises). Under Kahl's metanalysis, the same distribution is also predicted, because the metanalysis source forms were t-final stems, which would naturally not include stop-final bases. Both hypotheses therefore converge on the same prediction, and the test distinguishes both from the older "irregular extension" framing but not from each other.

Table 2. *Distributional test. Bare *-(e)ro- forms cluster on stop-final bases; t-bearing *-tero- forms cluster on bases ending in vowels, sonorants, and fricatives.*

Branch	Bare *-*(e)ro*- on stop-final bases	T-bearing *-*tero*- on non-stop bases
Sanskrit	*ádhara-* (*ṇ̥dh-*, dental asp.) *ápara-* (**ap-*, labial stop) *ávara-* (**au-*, glide)†	*uttara-* (**ud-*, dental → gemination)
Avestan	*upara-* (**up-*, labial stop) *aδara-* (**adh-*, dental) *apara-* (**ap-*, labial stop)	*fra-tara-* (**pro-*, vowel)
Latin	*super* (**(s)up-*, labial stop) *inferus* (**ṇ̥dh-*, dental)	*inter, intra* (**en-*, nasal) *exter, extra* (**eks-*, fricative) *ultra* (**ol-*, liquid) *citra* (**ḱi-*, vowel) *contra* (**kom-*, nasal)
Greek	*hupér* (**(s)up-*, labial stop)	*próteros* (**pro-*, vowel) *éntera* (**en-*, nasal) *deksiterós* (**deḳsi-*, vowel)

† *ávara-* (**au-*, glide-final) is the weakest match: the base ends in a glide rather than a stop. We treat it as a lexicalized relic; it does not falsify the prediction.

7.2 Results and exceptions

Latin shows the most extensive paradigm: seven of seven inherited forms match the prediction. Two Latin forms do not fit: *subter* "below" (base **(s)up-*, labial stop) and *propter* "near" (base **prop-*, labial stop). Both are Latin-internal formations modeled on existing **-tero-* words (*subter* on *inter*, *propter* on *praeter*), and Lewis & Short and de Vaan (2008) treat them accordingly. They illustrate what unconstrained analogical extension looks like once the original conditioning rule has ceased to be phonologically active: Latin adds *-ter-* freely to stop-final bases because the OCP constraint that conditioned the original alternation no longer applies. This is itself evidence for the reversal hypothesis: the inherited paradigm shows systematic conditioning, while the later Latin extensions do not.

Two Greek forms commonly cited in this connection are etymologically opaque and fall outside the test. *Hústeros* "later" derives either from **ud-(t)ero-* or **uds-tero-* (Beekes 2010: "uncertain"); *asterós* "left" may derive from a superlative base **ar-isto-* with appended **-(e)ro-*. Neither constitutes a clean test case.

The pronominal possessive paradigm is the test's failure mode, and we acknowledge it openly. The first- and second-plural possessives show the predicted alternation breaking down across branches: Latin *noster, vester* have **-tero-* on sibilant-final bases; Gothic *unsar, izwar* have bare **-(e)ro-* on the same kind of bases. This is not predicted by any phonological conditioning rule. The defense is that the pronominal possessives are a small functional category that became generalized branch-internally once the original conditioning rule had ceased to operate — Latin generalizing **-tero-*, Germanic the bare **-(e)ro-*, Celtic preserving both. The same dynamic explains the inner-Latin exceptions.

7.3 What the test shows, and what it doesn't

Run on the inherited PIE positional adjective paradigm, the prediction is matched in approximately 19 of 19 etymologically transparent cases across Sanskrit, Avestan, Latin, and Greek. The pattern is not analogically irregular. The inherited paradigm shows systematic phonological distribution.

What the test does not show is which of the two competing accounts is correct. The framework's reversal predicts the systematic distribution from the OCP; Kahl's metanalysis predicts the same distribution from the source-form environments. Both are compatible with the data. The contribution of the test is therefore narrower than earlier drafts of this paper claimed: it argues against the "irregular extension" framing of Szemerényi, Sihler, Wackernagel and Debrunner, and Schwyzer, but does not by itself decide between the framework and Kahl. The framework's advantage over Kahl is etymological: the framework explains why the *t-* was there in the first place, as the initial consonant of the hand-noun **der-* devoiced in suffix-initial position. Kahl's metanalysis leaves the source of the *t-* open.

8. The Anatolian distribution argument

Of all the distributional facts in the IE contrastive system, the Anatolian asymmetry is the most informative. Anatolian preserves the bare **-(e)ro-* in lexicalized formations (*kattera-*) but shows no productive **-tero-* whatever. The productive contrastive function is taken over by **-eti-Ho-* (Hittite *-ezziya-*). The standard view treats this as an arbitrary inheritance: Anatolian "happens" to have generalized one alternant and not the other. The reversal view converts this coincidence into a structural prediction.

The reasoning is as follows. If the t-bearing **-tero-* is original at the pre-PIE stage, and the bare **-(e)ro-* arises by cluster simplification in lexicalized formations, then at any given stage of the proto-language we expect: productive **-tero-* derivations on stems where no cluster simplification applies, and lexicalized **-(e)ro-* relics in formations where the simplification has applied. The productive system is always **-tero-*; the bare alternants are always relics.

Now consider the Anatolian split. If Anatolian split off from PIE early — as the Indo-Anatolian hypothesis maintains, with positive evidence from morphology, vocabulary, and verb structure (Kloekhorst 2008: 7–11; Melchert 2014; Kloekhorst & Pronk 2019) — then it represents a stage of the proto-language earlier than PNIE. At that stage, the productive **-tero-* system may not yet have generalized across the morphology. What had stabilized were the lexicalized formations on the most frequently-occurring local particles — like **ḱm̥t-* "below" — where the simplification had already produced bare forms. These bare forms, lexicalized as adjective stems, were available for inheritance into daughter branches.

Under this picture, Anatolian inherits exactly what it shows: bare relics in lexicalized contrastive formations (*kattera-*) but no productive **-tero-* system. The productive system develops post-Anatolian, in PNIE, when the suffix generalizes across the contrastive paradigm in non-cluster environments. Sanskrit, Greek, Italic, Tocharian, and Germanic all inherit the productive system because they descend from the post-Anatolian PNIE. Anatolian does not, because its split predates the generalization. Under the standard view, the Anatolian distribution is unmotivated. The framework converts it from a coincidence into a prediction.

9. Cross-cultural evidence: HAND > BINARY-CONTRASTIVE

The framework's morphological claim has a typological prediction: if pre-PIE **der-* "hand" grammaticalized into a binary-contrastive marker, similar pathways or at least the component links should be attested in other languages. The honest report: the direct pathway HAND > BINARY-CONTRASTIVE is not catalogued in Heine & Kuteva (2002), the standard reference on grammaticalization pathways. What is documented are HAND > AGENT MARKER, HAND > LOCATIVE/GENITIVE/POSSESSIVE, HAND > FIVE, and HAND > MANNER. None of these is the binary-contrastive marker we are claiming. The closest is HAND > LOCATIVE, which forms one of the two component links we need.

9.1 Component links and typological parallels

HAND > SIDE is documented in P"urhepecha (the suffix *-k'u* from *jak'i* "hand" marks the locative on body parts), Mordvin (*ked'* "hand" shows partial application of the body-part-to-spatial pathway), and the Mayan languages (Kaqchikel, Tz"utujil, K"iche', Tzotzil use body-part terms productively as relational nouns expressing spatial locations). SPATIAL > BINARY-CONTRASTIVE is documented in Tungusic and Uralic.

9.2 Kahl's typological parallels

Kahl (2024) — discussed in §4 — documents typological parallels for binary-contrastive morphemes that bear directly on the framework.

In **Uralic**, Proto-Uralic **-mpV* is reflected in Finnish as the productive comparative (*iso* "large" — *iso-mpi* "larger"). But beside this productive comparative function, Finnish preserves "synchronically unusual binary-contrastive forms" (Kahl 2024: 12, citing Hakulinen 1957: 74–76) — the relative pronoun *jompi* and its interrogative equivalent *kumpi* "which of two." The same morpheme serves both the productive comparative function and a fossilized binary-contrastive function — exactly the functional pairing the reversal hypothesis predicts for PIE **-tero-*.

In **Saami**, South Saami *gåabpa* and North Saami *goabbá* "which of two" are formally comparatives (Kahl 2024: 12). In some Saami varieties the parallel with Indo-Iranian extends further: the apparent superlative *guhtemusš* "which of several" corresponds to Vedic *katamá-*, Avestan *katāma-* "which of several." This fine-grained correspondence is not plausibly attributable to language contact.

In **Tungusic**, the suffix Proto-Tungusic **-dımar* / Proto-Northern Tungusic **-tmAr* appears both as an optional comparative marker and as a binary-contrastive marker. Kahl notes that "it seems that the binary-contrastive function is older and was secondarily drawn into comparative constructions" (Kahl 2024: 13). This is precisely the diachronic ordering the framework proposes for PIE: binary-contrastive **-tero-* (the original PIE function) generalized to gradational comparative in the Greek and Indo-Iranian innovation. The Tungusic parallel provides an independently attested case of the same diachronic development.

9.3 The English idiom

There is one piece of cross-linguistic evidence for HAND used directly for binary contrast that does not require typological reconstruction: the English idiom "on the one hand … on the other hand." When a speaker says "on the one hand the evidence supports X; on the other hand it supports Y," HAND is doing the work of marking the two terms of a binary opposition. The construction is alive, productive, and unmistakable. It is not in Heine & Kuteva's database, presumably because it is idiomatic rather than fully grammaticalized into a bound morpheme. But the function it performs is exactly the function we are claiming for pre-PIE **der-* in its grammaticalized form: HAND as the pivot for binary contrast.

9.4 The convergent case

The argument rests on nine distinct lines of evidence. Rather than narrating each, we collect them in Table 3.

Table 3. *Nine reasons *-tero- is semantically consistent with an original hand meaning.*

#	Reason	Evidence and source
1.	The lateralization domain is hand-related	The suffix productively forms hand-side terms across multiple branches: Latin *dexter*, Greek *deksiterós*, Old English *swiþro* — "right (hand)" — right side (§3).
2.	Body parts lexicalize from the spatial adjective	The neuter of **h₁en-tero-* "inner" lexicalizes as "entrails" independently in three branches: Latin *intestina*, Greek *éntera*, Sanskrit *āntrá-* (§2). Body-part lexicalization confirms the body-part orientation of the suffix.
3.	The paradigm matches the predicted grammaticalization output	Almost every PIE local particle takes **-tero-*; the result names the spatial contrastive of the particle (Table 1). This is precisely the output a HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE grammaticalization would predict.
4.	The binary-contrastive function uses a hand-pivot in living language	**-tero-* marks "which of two" (Latin *uter*, Sanskrit *katára-*, Greek *póteros*). English "on the one hand … on the other hand" uses HAND as the explicit binary-contrast pivot in living speech (§9.3).
5.	Component pathways are independently attested	HAND > SIDE in P"urhepecha, Mordvin, Mayan (Heine & Kuteva 2002). SPATIAL > BINARY-CONTRASTIVE in Tungusic, Uralic, Saami (Kahl 2024).
6.	The diachronic ordering is	Tungusic shows BINARY-CONTRASTIVE > COMPARATIVE — exactly the ordering PIE shows, with gradational comparison as the late Greek/Indo-Iranian innovation (Handbook vol. 3;

	independently attested	Milizia 2020).
7.	*Ambidexter* presupposes the bilateral norm	Latin *ambidexter* parses as *ambi-* + **deḱs-* + *-i-* + **-tero-* = "on-both-sides + right-handed + CONTRASTIVE." The compound is only meaningful if **-tero-* normally marks one of two bilateral sides (§3).
8.	The Anatolian distribution becomes a prediction	Anatolian preserves a lexicalized bare relic (*kattera-*) but no productive **-tero-*. Under the framework, this follows from the chronology of the Anatolian split. Under the standard view, it is unmotivated (§8).
9.	Kahl (2024) independently supports the locatival **-r-*	Kahl (§2.3.1) independently reconstructs **-(e)ro-/-er* as a locatival morpheme. The framework reads **-tero-* as **de-* (hand) + **-r-* (locative) + **-o-* (thematic). Kahl and the framework agree on **-r-* being locative; the disagreement is only on whether the *t-* is metanalyzed or original (§4.2).

No single line forces the framework. Each could be accommodated under an alternative account. What we claim is that the convergence is unusual: the framework explains nine separate observations from a single etymological source, while the standard view leaves each as an unconnected fact about the morphology. The framework's value is in the unification.

10. Pointers: **terh$_2$-* and the broader **der-* family

Two broader extensions of the **der-* framework — the directional verb **terh$_2$-* "to cross over" (Latin *trāns*, Sanskrit *tirás*, Hittite *tarḫu-*) and the maintenance verb **dher-* "to hold" with its reflex in Sanskrit *dharma* — are flagged here as terrain the framework touches but does not develop in this paper. Both are consistent with the framework's proposed hand-semantics network; both require their own paper-length treatment. The canonical discussion, including the *dharma* etymology and the semantic chain HAND > HOLD > MAINTAIN > COSMIC ORDER, is in Paper 3 §7, to which we point the reader.

11. Future research

This paper has restricted itself to the contrastive **-tero-*. Three morphological extensions that the framework also touches will each require their own treatment.

The kinship suffix **-(h$_2$)ter-* — in **māter-* "mother," **ph$_2$ter-* "father," **b^{h}rāter-* "brother," **d^{h}ugh$_2$ter-* "daughter" — shares the t-r structure of the contrastive **-tero-*. Whether it descends from the same pre-PIE etymon, perhaps via a possessive/relational pathway (HAND > GENITIVE/POSSESSIVE in Heine & Kuteva 2002) and a kinship-classifier development, is an open question. Latin *matertera* "maternal aunt" — a Latin-internal extension of **-tero-* to a kinship term — hints at the pathway. Aufderheide & Keydana (2016), who argue that the *i* in Sanskrit *pitár-* "father" is a synchronic repair rather than a laryngeal reflex, provide relevant starting-point material for the future paper.

The agent suffix **-ter-/-tor-* — in Greek *dó-tor*, Latin *da-tor*, Sanskrit *dā-tār-* — similarly shares the t-r structure. The cross-linguistic pathway HAND > AGENT MARKER is one of the well-documented HAND grammaticalizations (Heine & Kuteva 2002: 167–168). Whether the PIE agent suffix descends from the same pre-PIE etymon is the natural extension of the present paper.

The heteroclitic **-r/n-* body-part stems (**úód-r̥/*uéd-n̥s* "water," **peh$_2$-ú̥r/*peh$_2$úen-* "fire," **yekw-r̥/*yekw-en-* "liver") connect more directly with Paper 3, which develops a compositional analysis of the etymon with **-r-* as an inherited locative formative.

12. Remaining problems

Three problems deserve flagging, none of which we regard as fatal.

First, the directionality question is genuinely contested. The standard view and Kahl's more sophisticated metanalysis hypothesis are both consistent with the distributional data. The framework provides independent reasons for the reversal — the etymological motivation of the hand-noun source — but the distributional test of §7 does not by itself decide between reversal and metanalysis. The reversal predicts the systematic distribution from the OCP rule; metanalysis predicts the same distribution from the source-form environments. Both are compatible with the data.

Second, the proposed grammaticalization pathway from HAND to the contrastive function is supported by typological parallels in the component links but does not have direct PIE-internal documentation. The *deksiterós/dexter* compounds preserve the lateralization function transparently, but the further step — from lateralized relational form to general spatial-direction marker on local particles — is reconstructed rather than directly attested.

Third, the timing of the various grammaticalizations relative to PIE-internal sound changes is not fully worked out. The cluster simplification is placed at pre-PIE/early PIE, before the Anatolian split, but a fuller chronology remains to be developed. Each of these is a place where further work is needed; none refutes the proposal.

13. Conclusion

The morphological argument rests on a paradigm-level observation: the PIE contrastive suffix **-tero-* is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Every major PIE local particle has a **-tero-* derivative naming the spatial-contrastive of that particle, from Latin *citra/ultra/intra/extra/contra* to Sanskrit *ádhara-/uttara-/ápara-/ántara-* to Avestan and Old Persian *fratara-/apatara-/upatara-* to Hittite *kattera-*. The Handbook states explicitly that this spatial-particle function is the original PIE function; the gradational "more X" use is a Greek and Indo-Iranian innovation.

The framework of Paper 1 — pre-PIE **der-* "hand, open palm" — provides exactly the etymological motivation the standard treatment lacks. A hand-noun grammaticalizing through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE is exactly what one would expect to find as the systematic spatial-direction marker on local particles. The mainstream consensus that **-tero-* marks bilateral contrast is not the framework's claim; it is the established starting point. The framework answers what the mainstream leaves open: where the bilateral character came from.

The Latin compound *ambidexter* presupposes the bilateral norm by negating it; Hittite *kattera-* anchors the OCP phonology (suffix-initial deletion confirmed over root-final deletion by the geminate *-tt-*); the distributional test argues against the "irregular extension" framing; and the Anatolian distribution — bare relic but no productive **-tero-* — is a structural prediction of the framework, unmotivated on the standard view. Nine convergent reasons are collected in Table 3.

We do not claim the hypothesis is correct in all its details. The directionality is contested; the grammaticalization pathway is reconstructed rather than directly attested; the pre-PIE chronology is open. We do claim that the hypothesis is internally consistent, provides a unified explanation where the standard treatment provides only description, and deserves examination as an alternative reading of a morphological landscape that has been catalogued but not etymologized for over a century.

14. Probability estimate

Following the practice established in Paper 1, we close with an explicit probability assessment framed conditionally.

Conditional on the central claim of Paper 1 being correct — i.e., that pre-PIE had a body-part word **der-* meaning "hand, open palm" — Claude's estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately 35–45%. The §7 distributional test and the recognition that Kahl's locatival origin for **-(e)ro-* supports the framework's reading of **-r-* raise the conditional estimate above what earlier drafts implied.

The marginal probability — unconditioned, integrating over the probability that Paper 1 is also correct — is approximately the product: 15–20% (Paper 1) × 35–45% (Paper 2 conditional on Paper 1) ≈ 5–9%. This substantially lower marginal figure represents the cumulative ambition of the framework more accurately than earlier framings implied. We continue to regard the central claim as more likely wrong than right, and also to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted. These two judgments are compatible.

15. Authorship and contributions

This is the seventh version of Paper 2 in the *der-* project. It builds on Paper 1 (Chavis & Claude 2026a) and is developed further in Paper 3 (Chavis & Claude 2026c). We continue the same co-authorship arrangement.

Chavis's contributions. The substantive intellectual direction originates with Chavis. The hypothesis that *der- "hand" underlies the contrastive *-tero- was first articulated by Chavis, as was the slogan "the *ter* in **-tero-* is hand." Chavis identified *ambidexter* as the morphologically transparent witness and directed the investigation toward the bilateral-contrast argument developed in §3. Chavis insisted that the phonological argument rely entirely on existing PIE rules rather than positing a new sound law — a constraint that led to the engagement with Byrd (2015) and the identification of footnote 161 as the wiggle room. The distributional test in §7, the Anatolian distribution argument in §8, and the engagement with Kahl (2024) were all directed by Chavis. The comprehensive rewrite that produced v7 was directed by Chavis in response to accumulated structural and prose concerns across the project's arc review.

Claude's contributions. Claude's contributions are primarily in source synthesis, exposition, and analytical development from Chavis's structural direction. Claude searched and read the relevant published literature (Lundquist & Yates 2017; Kulikov, Dupraz, Milizia, and Malzahn & Fellner in Keydana et al. 2020; Kloekhorst 2008; Byrd 2015; Melchert 1994 and 2017; Heine & Kuteva 2002; de Vaan 2008; Kahl 2024; Lipp 2009; Lipp 2017; Aufderheide & Keydana 2016). Claude identified Byrd's footnote 161 as the explicit recognition of suffix-initial deletion; documented the Latin *-trā/-trō* paradigm and the Avestan/Old Persian "oldest nucleus" statement from Milizia; developed the distributional test cases; surfaced Kahl's typological parallels (Uralic, Saami, Tungusic) for §9; and proposed the Anatolian distribution argument of §8. The probability estimate in §14 is Claude's own.

Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project across conversations is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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